Inputs and Outputs of Major Codes in a Stellarator Design Workflow

stellarator_workflow repository draft

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Abstract

This document isolates the software contracts that appear in a modern stellarator design loop. For each major code family it lists the main input artifacts, the main output artifacts, the subset of outputs that are normally handed to the next code in the pipeline, and the subset of outputs that are commonly used as optimization or design objectives. The emphasis is on the practical interfaces that connect VMEC or DESC equilibrium calculations to Boozer transforms, neoclassical solvers, gyrokinetic solvers, and global transport tools [1, 2, 4, 14, 15, 18].

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1 Purpose and conventions

The goal here is narrower than the main workflow manuscript. This document is about *interfaces*: what each code expects to read, what it writes or returns, which outputs are consumed by the next stage, and which outputs are typically minimized, constrained, or monitored during stellarator optimization.

Three conventions are used throughout:

- "Input" means either an on-disk file accepted by the public CLI or a public in-memory object accepted by the documented Python or JAX API.
- "Output" means either a persisted artifact such as `wout_*.nc`, `boozmn_*.nc`, `sfincsOutput.h5`, or a documented result object returned by a public API.
- "Optimization objective" means a quantity that is commonly included directly in an objective function or used as a screening metric, even when it is *not* passed to the next code in the transport loop.

The repository snapshots under `_sources/` were used to verify the representative NetCDF and HDF5 schemas quoted below, so the tables are tied to actual artifacts rather than only to README summaries [7, 10–13, 16, 24–31].

2 Executive summary

2.1 Pipeline hand-offs

Table 1: What each stage usually hands to the next stage.

Code family	Main output artifact	Next code(s)	Subset usually handed forward
VMEC / vmec_jax	`wout_*.nc` or in-memory wout-like object	BOOZ_XFORM, booz_xform_jax, GX, Trinity3D, NEOPAX	equilibrium spectra and profiles, especially `rmnc`, `zmns`, `lmns`, `bmnc`, `bsubumnc`, `bsubvmnc`, `iotas`, `phi`, `phipf`, geometry scalars, and pressure / current profiles
DESC	`*.h5` equilibrium family, plus optional exported VMEC or Boozer files	MONKES directly; otherwise the same downstream codes as VMEC after export	surface and equilibrium coefficients, pressure / iota / current profile parameters, and any exported `wout`- or `boozmn`-compatible file
BOOZ_XFORM /			

Code family	Main output artifact	Next code(s)	Subset usually handed forward
booz_xform_jax	`boozmn_*.nc` or in-memory Boozer object	NEO, NEO_JAX, SFINCS, sfincs_jax, MONKES, NEOPAX	Boozer field spectrum and radial profiles, especially `bmnc_b`, `ixm_b`, `ixn_b`, `iota_b`, `buco_b`, `bvco_b`, `phi_b`, `phip_b`, and the selected surface list
NEO / NEO_JAX	`neo_out.*`, `neolog.*`, optional current / diagnostic files, or in-memory surface results	usually no direct transport consumer	usually none; optional current-related outputs may be fed into outer current or equilibrium updates
SFINCS / sfincs_jax	`sfincsOutput.h5`, optional `.npy` state or transport-matrix dumps	Trinity3D or an outer equilibrium / transport loop	most often `particleFlux_vm_rN` and `heatFlux_vm_rN`; if `includePhi1` is enabled in the Trinity3D adapter, the hand-off becomes `particleFlux_vd_rN` and `heatFlux_vd_rN`; bootstrap-current and `E_r` information may also be reused
MONKES	in-memory `Dij`, `f`, `s`, or an HDF5 monoenergetic database	NEOPAX	for today's NEOPAX reader the effective hand-off is the reduced database: `D11`, `D13`, `D33`, `Er`, `Er_tilde`, `drds`, `rho`, and `nu_v`
GENE / GENE-3D	installation-specific output files or restart / diagnostic datasets	Trinity3D-like profile solvers, validation workflows	usually species heat and particle fluxes; in linear studies only γ and ω may be kept
GX	`run_name.out.nc`, `run_name.nc`, `run_name.big.nc`, `run_name.restart.nc`	Trinity3D	steady estimates of particle and heat flux, read in the present Trinity3D adapter from either `Diagnostics/ParticleFlux_st` and `Diagnostics/HeatFlux_st` or from `Fluxes/pflux` and `Fluxes/qflux`; zeta-resolved stellarator runs can also use `ParticleFlux_zst` and `HeatFlux_zst`
SPECTRAX-GK	in-memory diagnostics and optional CSV / plot outputs	outer optimization or a transport wrapper	the natural hand-off is the same turbulent contract as GX: `gamma`, `omega`, heat-flux traces, and particle-flux traces

Code family	Main output artifact	Next code(s)	Subset usually handed forward
Trinity3D	`*.nc`, `*.bp`, `*.log.npy`, `*.out`	VMEC or DESC in the next outer iteration	updated density and temperature profiles, hence an updated pressure profile; in some workflows bootstrap-related current information is also recycled
NEOPAX	in-memory `Lij`, `Gamma`, `Q`, `Upar`; solved profile states; optional HDF5 outputs	VMEC or DESC in the next outer iteration	updated `E_r(r)`, neoclassical fluxes, and pressure or current-related profiles from the reduced transport solve

2.2 Outputs commonly used as objectives

Table 2: Outputs commonly used as stellarator optimization or design objectives.

Code family	Outputs commonly used in objectives	Also a direct transport input?
VMEC / vmec_jax	aspect ratio, volume, β, rotational-transform targets, magnetic-well / Mercier-related metrics, boundary smoothness, equilibrium existence / residual quality	indirectly; the equilibrium itself is the upstream state for later codes
DESC	quasisymmetry error, geometry constraints, aspect ratio, volume, rotational transform, force-balance residuals, differentiable proxy objectives	indirectly; exported equilibria feed the rest of the workflow
BOOZ_XFORM / booz_xform_jax	non-target Boozer harmonics, symmetry-breaking spectrum, mirror terms, quasisymmetry error in Boozer coordinates	usually no; mainly a geometry diagnostic and interface layer
NEO / NEO_JAX	ϵ_{eff}, class-resolved effective ripple, trapped-particle screening metrics, current proxies	usually no
SFINCS / sfincs_jax	neoclassical particle flux, heat flux, bootstrap current, ambipolar E_r, Φ_1, transport matrices	yes
MONKES	monoenergetic D_{ij} coefficients and their sensitivity to radius, collisionality, and E_r	yes, through reduced neoclassical models such as NEOPAX
GENE / GENE-3D	linear growth rates and frequencies, nonlinear heat and particle fluxes, transport stiffness	nonlinear fluxes yes; linear rates usually no
GX	linear γ and ω, nonlinear heat flux, particle flux, free energy, stiffness	yes for heat and particle flux
SPECTRAX-GK	`gamma`, `omega`, `heat_flux`, `particle_flux`, `Wg`, `Wphi`	yes once wrapped into a transport loop
Trinity3D	`Pfus_MW`, `Qfus`, `Paux_MW`, transport-consistent $n(r)$, $T(r)$, and β	it is already the global transport stage

Code family	Outputs commonly used in objectives	Also a direct transport input?
NEOPAX	ambipolar E_r, bootstrap current, toroidal current, neoclassical fluxes, reduced-order profile predictions	yes

3 Code-by-code reference

3.1 VMEC and vmec_jax

Role. VMEC solves the fixed-boundary ideal-MHD equilibrium; `vmec_jax` preserves essentially the same physical contract while exposing a JAX-native, differentiable API [13, 14, 31].

Primary inputs. The canonical VMEC input is `input.NAME`. The core design variables are the boundary Fourier coefficients `RBC(m,n)` and `ZBS(m,n)`, plus pressure coefficients `AM` or the auxiliary pressure arrays `AM_AUX_*`. Either rotational transform or current is prescribed through `AI`, `AC`, and their auxiliary arrays. The magnetic scale is set by `PHIEDGE`. Resolution and solver controls include `NS`, `MPOL`, `NTOR`, iteration budgets, and tolerance arrays. The JAX version accepts the same input file and additionally supports in-memory resume state and solver options.

Primary outputs. The main output is `wout_*.nc`. Verified example files contain scalar geometry and convergence quantities such as `aspect`, `Aminor_p`, `Rmajor_p`, `volume_p`, `betatotal`, `b0`, `volavgB`, `fsqr`, `fsqz`, and `fsql`. The radial-profile outputs include `presf`, `pres`, `phi`, `phipf`, `chi`, `chipf`, `iotas`, `iotaf`, `q_factor`, `jcuru`, `jcurv`, `buco`, and `bvco`. The spectral geometry is stored in arrays such as `rmnc`, `zmns`, `lmns`, `bmnc`, `gmnc`, `bsubumnc`, `bsubvmnc`, `bsubsmns`, `currumnc`, and `currvmnc`. The JAX API also returns a `FixedBoundaryRun` object containing the solver state, flux profiles, and a wout-like in-memory representation.

Subset handed to the next code. BOOZ_XFORM and `booz_xform_jax` need the full equilibrium spectrum and profiles encoded in `wout_*.nc`. GX, Trinity3D, and NEOPAX geometry readers also consume `wout`-level data rather than only high-level scalars, because they need field-line geometry, the rotational transform, and surface metrics.

Subset used as objectives. Common direct objectives are aspect ratio, volume, β, target $\iota(s)$, and equilibrium-quality metrics. The residual scalars `fsqr`, `fsqz`, and `fsql` are mainly QA signals rather than physics design objectives.

3.2 DESC

Role. DESC is a differentiable pseudo-spectral equilibrium and optimization suite [3–5, 23, 24].

Primary inputs. DESC accepts an input file or Python configuration specifying the surface representation, field period, toroidal flux, pressure profile, and either iota or current profile, together with spectral resolution, continuation settings, and optimizer options. It can also import VMEC-like inputs and outputs.

Primary outputs. The standard serialized output is `*.h5` or `*.pkl`. Example HDF5 equilibrium families contain groups such as `_equilibria/*` with the spectral coefficients `_R_lmn`, `_Z_lmn`, `_L_lmn`, surface and axis descriptions, toroidal flux `_Psi`, and saved pressure and iota profile parameters under `_pressure/_params` and `_iota/_params`. DESC can also export VMEC-style or BOOZ_XFORM-compatible files through its VMEC utilities.

Subset handed to the next code. If the next code expects VMEC input, the hand-off is an exported `wout`. If the next code expects Boozer data, DESC can provide a `boozmn`-style file or a direct in-memory field object. MONKES can consume DESC-derived field objects directly, while VMEC-centric tools usually rely on exported `wout`-compatible data.

Subset used as objectives. DESC is itself commonly the optimization host, so quasisymmetry error, aspect ratio, volume, iota targets, boundary regularity, and force-balance residuals are often objectives. Once coupled to GX or another turbulent model, turbulent transport metrics become part of the same objective stack.

3.3 BOOZ_XFORM and booz_xform_jax

Role. These tools transform a VMEC-like equilibrium into Boozer coordinates [2, 12, 26].

Primary inputs. The legacy route is a VMEC `wout_*.nc` file plus an `in_booz.*` control file specifying resolution and the list of radial surfaces to compute. The JAX version also accepts an in-memory wout-like object.

Primary outputs. The standard file output is `boozmn_*.nc`. Verified examples contain `iota_b`, `pres_b`, `beta_b`, `phip_b`, `phi_b`, `bvco_b`, `buco_b`, `jlist`, mode lists `ixm_b` and `ixn_b`, and Fourier arrays `bmnc_b`, `rmnc_b`, `zmns_b`, `pmns_b`, and `gmn_b`. The JAX API also holds Boozer I and G profiles and the selected radial grid in memory.

Subset handed to the next code. NEO and NEO_JAX need the Boozer spectrum and its radial profiles. SFINCS and `sfincs_jax` use the same Boozer geometry when running in a Boozer-based geometry mode. MONKES and NEOPAX use the Boozer spectrum through direct field or file readers.

Subset used as objectives. The most common design outputs are the non-target Boozer harmonics in `bmnc_b`, or equivalent symmetry-breaking measures built from them. These are geometry objectives, not state variables advanced by a transport solver.

3.4 NEO and NEO_JAX

Role. NEO computes effective ripple and related trapped-particle diagnostics from Boozer data [21, 22, 25, 27].

Primary inputs. The standard inputs are a `boozmn_*.nc` file and a control file such as `neo_param.ext` or `neo_in.ext`. The control file sets the surface list, angular resolution (`theta_n`, `phi_n`), Fourier cutoffs, Monte Carlo controls, accuracy targets, and switches for current calculation and optional extra file dumps. NEO_JAX also exposes a direct Python or JAX API that returns structured result objects.

Primary outputs. The legacy outputs are `neo_out.*` and `neolog.*`, plus optional `neo_cur.*`, `current.dat`, `conver.dat`, `diagnostic.dat`, `diagnostic_add.dat`, and `diagnostic_bigint.dat`. The principal physics quantities are `epstot` ($= \epsilon_{\text{eff}}^{3/2}$), `epspar`, `reff`, `iota`, `b_ref`, `r_ref`, `ctrone`, `ctrtot`, `bareph`, `barept`, and `yps`. In NEO_JAX these appear in result objects under names such as `epsilon_effective` and `epsilon_effective_by_class`.

Subset handed to the next code. Usually none. In most workflows NEO is a screening or optimization branch, not a transport branch. If `CALC_CUR=1`, current-related outputs can be fed into a broader outer loop, but that is not the default Trinity3D hand-off.

Subset used as objectives. ϵ_{eff} and its class-resolved pieces are classic stellarator screening objectives. This is precisely the case in which a code's output is central to optimization but not central to profile evolution.

3.5 SFINCS and sfincs_jax

Role. SFINCS solves the radially local drift-kinetic equation for neoclassical transport in non-axisymmetric plasmas [15, 16, 20, 29].

Primary inputs. The main input is `input.namelist`. In current public interfaces the key groups are `general`, `geometryParameters`, `speciesParameters`, `physicsParameters`, `resolutionParameters`, `otherNumericalParameters`, `preconditionerOptions`, and optional `export_f` settings. Together they define the geometry source, radial coordinate, species charges and masses, $\hat{n}_s$, $\hat{T}_s$, their gradients, the collision model, electric-field terms, Φ_1 switches, and numerical resolution.

Primary outputs. The standard output is `sfincsOutput.h5`. Verified example files include geometry snapshots, copied input metadata, and the following reusable transport datasets:

- `particleFlux_vm_psiHat`
- `particleFlux_vm_psiN`
- `particleFlux_vm_rHat`

- `particleFlux_vm_rN`
- `heatFlux_vm_psiHat`
- `heatFlux_vm_psiN`
- `heatFlux_vm_rHat`
- `heatFlux_vm_rN`
- additional transport state: matching momentum-flux arrays, classical fluxes, `FSABjHat`, `FSABFlow`, `Phi1Hat`, and `transportMatrix`.

Optional exports include full-f, δf, and `*_vs_x` lineouts. The JAX interface additionally exposes in-memory result dictionaries and can write `.npy` state vectors or transport matrices from its CLI.

Subset handed to the next code. The present Trinity3D adapter reads `particleFlux_vm_rN` and `heatFlux_vm_rN` when `includePhi1` is off, and `particleFlux_vd_rN` and `heatFlux_vd_rN` when it is on. In other workflows `FSABjHat`, ambipolar electric-field information, and the transport matrix are also reused by an outer iteration.

Subset used as objectives. Neoclassical particle flux, heat flux, bootstrap current, ambipolar E_r, and Φ_1-related quantities are all common optimization metrics. Unlike ϵ_{eff}, many of these are also direct transport inputs.

3.6 MONKES

Role. MONKES is a fast JAX monoenergetic drift-kinetic solver [6, 7].

Primary inputs. The public API constructs a field object from a DESC equilibrium or an equivalent magnetic representation, then solves at a chosen radius with a species list, radial electric field, particle speed, and pitch-angle resolution. In database-building use, the same solve is repeated over collisionality and E_r grids.

Primary outputs. The core output is the monoenergetic transport matrix `Dij`, together with the perturbed distribution function `f` and source array `s`. In a database context these are stored as HDF5 arrays such as `D11`, `D13`, `D31`, `D33`, `rho`, `nu_v`, `Er`, `Er_tilde`, `Es`, and geometric conversion factors.

Subset handed to the next code. NEOPAX's public database reader currently consumes the reduced subset `D11`, `D13`, `D33`, `Er`, `Er_tilde`, `drds`, `rho`, and `nu_v`. The full distribution function `f` and source term `s` are useful for analysis but are not part of that hand-off.

Subset used as objectives. MONKES is often used to build cheap neoclassical surrogates. In that role the `Dij` coefficients themselves are optimization or screening targets.

3.7 GENE and GENE-3D

Role. GENE and GENE-3D provide high-fidelity gyrokinetic turbulence calculations in flux-tube and global settings [8, 9, 19, 32].

Primary inputs. The stable physics contract is clear even though the exact file syntax is not publicly standardized across installations: geometry derived from VMEC or Boozer coordinates, species densities and temperatures, radial gradients, collision parameters, electromagnetic switches, and numerical-grid settings. For stellarator flux-tube workflows the geometry usually passes through a stellarator interface layer before reaching GENE.

Primary outputs. The usual outputs are linear growth rates, real frequencies, eigenfunctions, and nonlinear species heat and particle fluxes together with spectra and time histories. The exact filenames are installation-dependent, so the important contract is the variable-level one rather than a specific disk layout.

Subset handed to the next code. For transport coupling, the critical hand-off is the turbulent flux vector. For screening, only linear γ and ω may be retained.

Subset used as objectives. Linear growth rates are often used for rapid pruning; nonlinear heat flux, particle flux, and stiffness are higher-fidelity design objectives.

3.8 GX

Role. GX is a GPU-native gyrokinetic code with a workflow that is especially suitable for repeated turbulent-transport evaluations inside design loops [10, 17, 18].

Primary inputs. GX is run with `gx run_name.in`. The input file controls geometry, species, domain, time stepping, diagnostics, restart behavior, and numerical resolution. Stellarator calculations can use VMEC-driven geometry or imported field-line geometry, and the diagnostics flags `omega=true` and `fluxes=true` determine whether growth-rate and flux diagnostics are written.

Primary outputs. Linear runs write `run_name.out.nc`. Nonlinear runs write `run_name.nc`. Saved-field diagnostics may also produce `run_name.big.nc`. Restart output uses `run_name.restart.nc`. Verified NetCDF outputs contain groups such as `Grids`, `Geometry`, `Diagnostics`, and `Inputs`. The public docs show linear growth-rate diagnostics under `Special/omega_v_time`. The present Trinity3D adapter reads flux data from `Diagnostics/ParticleFlux_st`, `Diagnostics/HeatFlux_st`, their zeta-resolved variants, or from the `Fluxes/pflux` and `Fluxes/qflux` groups when present.

Subset handed to the next code. Trinity3D uses steady estimates of the heat and particle fluxes at each radius. It obtains flux Jacobians by rerunning GX on perturbed gradients and finite differencing those outputs.

Subset used as objectives. Ion and electron heat flux are common direct objectives. Linear γ and ω are also used for screening. In turbulence-aware optimization, heat flux is both an objective and a transport input.

3.9 SPECTRAX-GK

Role. SPECTRAX-GK is a JAX-native gyrokinetic solver aimed at differentiability and rapid experimentation [30].

Primary inputs. The current public interface is config-driven. A TOML config specifies grid, geometry, physics toggles, and time integration. Geometry can be analytic or imported from a GX-style `*.eik.nc` file, including VMEC-derived geometry.

Primary outputs. The natural outputs are JAX-native result objects with `gamma`, `omega`, and time traces such as `gamma_t`, `omega_t`, `Wg_t`, `Wphi_t`, `Wapar_t`, `heat_flux_t`, and `particle_flux_t`. The nonlinear CLI can also save a CSV with compact columns for time, growth rate, frequency, free energy, and species-resolved heat and particle flux.

Subset handed to the next code. The natural downstream contract is the same one used by GX: turbulent heat and particle flux, and optionally linear γ and ω. In the current public repository this is primarily an optimization or validation interface rather than a fixed Trinity3D adapter.

Subset used as objectives. Growth rate, real frequency, turbulent heat flux, particle flux, and free energy are all exposed directly and are therefore natural differentiable objective components.

3.10 Trinity3D

Role. Trinity3D is a global transport solver that evolves density and temperature profiles and computes whole-device power metrics [1, 11].

Primary inputs. The main input is a TOML-style `*.in` file with groups such as `[grid]`, `[time]`, one or more `[[model]]` blocks, `[[species]]`, `[geometry]`, `[physics]`, and `[log]`. Geometry can be read from a VMEC `wout`. Turbulence and neoclassical models add their own inputs, for example `gx_template` and `gx_outputs` for GX coupling.

Primary outputs. Trinity3D writes NetCDF, ADIOS2, NumPy-log, and text-log outputs. Verified NetCDF outputs contain geometry groups with `B0`, `Btor`, `a_minor`, `R_major`, `area`, and `grho`; profile histories such as `n_e`, `T_e`, `n_H`, `T_H`; total and model-labelled fluxes `pflux_*` and `qflux_*`; gradients `aLn_*`, `aLT_*`; source histories `Sn_*` and `Sp_*`; and whole-device metrics such as `beta_vol`, `Paux_MW`, `Palpha_int_MW`, `Pfus_MW`, and `Qfus`.

Subset handed to the next code. The essential outer-loop hand-off is the updated profile state: $n_s(r)$, $T_s(r)$, and therefore $p(r)$. That pressure profile is what must be sent back to VMEC or DESC if the loop is to be closed self-consistently. When current evolution is modeled, bootstrap-related current information can also be recycled into the next equilibrium solve.

Subset used as objectives. `Pfus_MW`, `Qfus`, `Paux_MW`, β, and the final transport-consistent profiles are end-of-pipeline design metrics.

3.11 NEOPAX

Role. NEOPAX is a JAX framework for reduced neoclassical transport and profile evolution [28].

Primary inputs. Public examples build NEOPAX from three inputs: a VMEC `wout`, a Boozer `boozmn`, and a MONKES-like monoenergetic database. The database reader currently expects HDF5 datasets such as `D11`, `D13`, `D33`, `Er`, `Er_tilde`, `drds`, `rho`, and `nu_v`. The profile state can be loaded from NTSS-like HDF5 files with arrays such as `r`, `Er`, `Te`, `ne`, `Pressure`, `I_bs`, and related transport quantities, or assembled directly in memory.

Primary outputs. The core neoclassical API returns `Lij`, `Gamma`, `Q`, and `Upar`. Example HDF5 outputs range from a minimal `Er_Test.h5` containing `rho`, `Er`, and `Jboots`, to larger NTSS-style profile files containing arrays such as `Pressure`, `FluxNeo`, `FluxQe`, `FluxQi`, `AmbiFlux`, `J_bs`, `I_bs`, `I_tor`, `beta`, `Te`, `TD`, `Tt`, `ne`, `nD`, and `nHe`.

Subset handed to the next code. In an outer loop the practical hand-off is the reduced transport state: `Er(r)`, neoclassical fluxes, and any updated pressure or current profiles. NEOPAX also contains turbulence-coupling utilities, but the public examples center on the neoclassical reduced model.

Subset used as objectives. Ambipolar electric field, bootstrap current, toroidal current, neoclassical fluxes, and reduced-order profile predictions are all candidate objective components or screening metrics.

4 The two most important distinctions

Two distinctions repeatedly matter in stellarator workflow design:

1. Some outputs are *screening objectives only.* The clearest example is NEO's ϵ_{eff}: it is central to ranking candidate geometries, but it is usually not what Trinity3D advances in time.
2. Some outputs are *both objectives and state hand-offs.* The clearest examples are turbulent heat and particle flux from GX, GENE, or SPECTRAX-GK, and neoclassical flux from SFINCS. These quantities are often minimized in optimization and are also the direct inputs needed to evolve density and temperature profiles.

That distinction is what determines whether a future containerized black-box workflow needs a result only for ranking, or whether it needs a robust, machine-readable interface that another code must parse automatically.

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